

Early Prediction of Alzheimer's Disease Using Machine Learning

1. Executive Summary

This project aims to develop a machine learning model that predicts the likelihood of Alzheimer's disease onset or progression in patients using clinical, cognitive, and biomarker data. The results will support earlier diagnosis, more timely intervention, and better allocation of clinical resources for at-risk individuals.

2. Problem Statement

Background: Alzheimer's disease is a progressive neurodegenerative condition that is often diagnosed only after significant cognitive decline has occurred. Earlier identification of at-risk patients can improve treatment planning and quality of life, but current diagnostic approaches rely heavily on late-stage symptoms and costly specialist evaluation.

Objective: Build a predictive model that classifies patients by their risk of developing or progressing in Alzheimer's disease based on available clinical, demographic, cognitive assessment, and biomarker data.

Scope: Focus on structured patient-level data (e.g., demographics, cognitive test scores, MRI-derived measures, and genetic/biomarker indicators such as APOE status) across a defined patient cohort or public research dataset.

3. Data Sources

Primary Data: Alzheimer's Disease Dataset (Kaggle, Rabie El Kharoua) — 2,149 patients with demographic details, lifestyle factors, medical history, clinical measurements, cognitive and functional assessments, symptoms, and a diagnosis of Alzheimer's Disease.

Secondary Data: Patient records including demographic information, cognitive assessment scores (e.g., MMSE, CDR), MRI-derived brain volume measurements, and biomarker/genetic data (e.g., APOE4 status), sourced from a public research dataset such as ADNI (Alzheimer's Disease Neuroimaging Initiative) or OASIS (Open Access Series of Imaging Studies).

4. Methodology

Data Collection: Obtain and consolidate patient-level datasets from the chosen source(s), ensuring proper handling of any data use agreements and de-identification requirements.

Data Preparation: Clean and preprocess the data, including handling missing values, encoding categorical variables, normalizing numerical features, and addressing class imbalance between diagnosed and non-diagnosed cases.

Analysis Techniques: Exploratory data analysis, feature selection, and classification modeling using approaches such as logistic regression, random forests, gradient boosting, and neural networks. Model evaluation using accuracy, precision, recall, F1-score, and ROC-AUC, with particular attention to recall given the cost of false negatives in a medical context.

Tools: Python (using libraries such as pandas, scikit-learn, and optionally TensorFlow/PyTorch for deep learning approaches), with Excel or Jupyter notebooks for preliminary exploration and visualization.

5. Expected Outcomes

Accurate classification of patients by Alzheimer's risk level using available clinical and imaging data.

A validated predictive model that could support clinicians in prioritizing patients for further diagnostic workup.

Identification of the most influential features (e.g., cognitive scores, brain volume, genetic markers) driving prediction, to inform clinical understanding.

Recommendations for how the model could be integrated into a clinical screening workflow.

6. Risks and Challenges

Data quality and completeness issues, including missing values and small sample sizes, may limit model performance.

Class imbalance (fewer positive/diagnosed cases) could bias the model toward the majority class and require techniques such as resampling or class weighting.

Model predictions carry real-world clinical implications; false negatives and false positives both have meaningful consequences, so rigorous validation and cautious interpretation are essential.

Ethical and privacy considerations around the use of sensitive patient health data must be addressed throughout the project.

7. Conclusion

This project has the potential to provide meaningful value by supporting earlier identification of Alzheimer's disease risk. The predictive model and insights developed through this work could contribute to more timely clinical intervention, better-informed care planning, and a deeper understanding of the factors associated with disease onset and progression.